

explain difference generative vs discriminative modeling

# Generative vs Discriminative Models

## Core Difference

- **Generative:** Models the *joint probability*  $P(X, Y)$  - "How was this data generated?"
- **Discriminative:** Models the *conditional probability*  $P(Y|X)$  - "What's the boundary between classes?"

## Generative Models

**Goal:** Learn the underlying data distribution to generate new samples.

**What they do:**

- Learn  $P(X, Y) = P(X|Y) \times P(Y)$
- Can generate new data points
- Understand the data generation process

**Examples:**

- Naive Bayes
- Gaussian Mixture Models (GMM)
- Hidden Markov Models (HMM)
- Generative Adversarial Networks (GANs)
- Variational Autoencoders (VAEs)

**Pros:**

- Can generate synthetic data
- Handle missing data well
- Work well with unlabeled data (semi-supervised)
- Often need fewer training samples

**Cons:**

- Usually computationally heavier
- May model unnecessary details
- Often less accurate for pure classification

## **Discriminative Models**

**Goal:** Learn boundaries between classes for classification/prediction.

**What they do:**

- Learn  $P(Y|X)$  directly
- Focus on decision boundaries
- Don't care about data generation

**Examples:**

- Logistic Regression
- Support Vector Machines (SVM)
- Decision Trees/Random Forests
- Neural Networks (for classification)
- Conditional Random Fields (CRFs)

**Pros:**

- Typically better classification accuracy
- Computationally efficient
- Directly optimized for prediction
- Simpler for pure classification tasks

**Cons:**

- Cannot generate new data
- Need labeled data
- Less interpretable about data structure

## Analogy

**Generative:** Like learning to draw cats and dogs so you can recognize AND draw them

**Discriminative:** Like learning to distinguish cats from dogs by their features only

## When to Use Which?

| Use Generative When      | Use Discriminative When  |
|--------------------------|--------------------------|
| Need to generate data    | Only need classification |
| Have missing data        | Have ample labeled data  |
| Semi-supervised learning | Need highest accuracy    |
| Want data understanding  | Need fast predictions    |

## Mathematical View

- **Generative:**  $P(Y|X) = P(X|Y)P(Y) / P(X)$  (Bayes' rule)
- **Discriminative:** Directly models  $P(Y|X)$

**Key Insight:** All generative models can be used for classification, but discriminative models cannot generate data. Generative models are "more ambitious" - they solve a harder problem (modeling the entire data distribution), which sometimes comes at the cost of classification accuracy compared to models focused solely on discrimination.